

SIMATS ENGINEERING

COURSE: COMPUTER NETWORKS (CSA0711)

CSA0711— ASSESSMENT TOOL 1 — CO3 - AT 1

Assessment Tool 1 – Experiments | CO3 | 10 Marks

Course Outcome: CO3 – Analyze the performance of core network protocols such as TCP, UDP, HTTP and DNS under varying conditions.

This print-ready document contains Questions 1–4 from the supplied assessment PDF. Sample captured values are retained as sample values; replace them with your own Wireshark observations if required by the faculty.

Question

Capture packets during a TCP connection setup. Identify the three packets involved in the handshake (SYN, SYN-ACK, ACK). Demonstrate the role of each flag and how they establish a reliable connection.

1.1 Objective

To use Wireshark to capture live traffic during the establishment of a TCP connection, identify the SYN, SYN-ACK and ACK segments that make up the three-way handshake, and explain how the flags and sequence/acknowledgment numbers cooperate to establish a reliable, connection-oriented session between a client and a server.

1.2 Tools / Software Required

- Wireshark (packet capture and analysis)
- A host machine with an active Wi-Fi/Ethernet network interface
- A web browser or terminal utility such as curl/telnet to generate a fresh TCP connection

1.3 Procedure

1. Open Wireshark and select the active network interface.
2. Start a capture and apply the display filter: `tcp.flags.syn == 1 || (ip.addr == <server_ip> && tcp)`.
3. Open a web page or use curl to initiate a fresh TCP connection.
4. Stop the capture and locate the first three packets exchanged between the client and server.
5. Expand the Transmission Control Protocol section and inspect Flags, Sequence Number and Acknowledgment Number.
6. Right-click the first SYN packet and choose Follow → TCP Stream to isolate the connection.

1.4 Sample Captured Output

No.	Time	Source	Destination	Flags	Seq / Ack	Info
1	0.000000	192.168.1.10	93.184.216.34	SYN	Seq=0	49514 → 80 [SYN], Win=64240, MSS=1460
2	0.028417	93.184.216.34	192.168.1.10	SYN, ACK	Seq=0, Ack=1	Server acknowledges client SYN and sends its SYN
3	0.028562	192.168.1.10	93.184.216.34	ACK	Seq=1, Ack=1	49514 → 80 [ACK], Win=64240

Sample Captured Output: